

Introduction to numerical methods - 12 -

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Green's function

Fourier transform

$$\tilde{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(\omega) e^{i\omega t} d\omega$$

Derivation

$$\frac{dH(t)}{dt} = \frac{d}{dt} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{F}(\omega) e^{i\omega t} d\omega \right] =$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{F}(\omega) i\omega \cdot e^{i\omega t} d\omega$$

$$\mathcal{F} \left[\frac{dH}{dt} \right] = i\omega \tilde{F}(\omega)$$

Usage example:

$$\frac{dx}{dt} + kx = \overset{\text{drive}}{g(t)} \quad \leftarrow \text{FT}$$

$$i\omega \tilde{x} + k\tilde{x} = \tilde{g}$$

$$\tilde{x} = \frac{\tilde{G}(\omega)}{i\omega + k}$$

Green's function: linear system

$f_i(t)$ input $x_i(t)$ output

$$\left. \begin{aligned} f_1(t) + f_2(t) &\rightarrow x_1(t) + x_2(t) \\ a \cdot f(t) &\rightarrow a \cdot x(t) \end{aligned} \right\} \text{linear system}$$

Let us have a driving function $f(t)$
what is the response of the linear system.

Every instance of $f(t')$ can in principle contribute to $x(t)$. The general linear relation:

$$(1) \quad x(t) = \int_{-\infty}^{\infty} G(t, t') f(t') dt'$$

$\hookrightarrow$ Green's function, effect of drive at t' on the output at time t

(2) Causality: obviously only past events can affect the present
(this is missing if $t \rightarrow y$)
time space

$$x(t) = \int_{-\infty}^t G(t, t') f(t') dt'$$

(3) Time invariance: we assume that the system does not change in time (the diff. eq. remains the same)

Only the time difference matters

$$\tau := t - t'$$

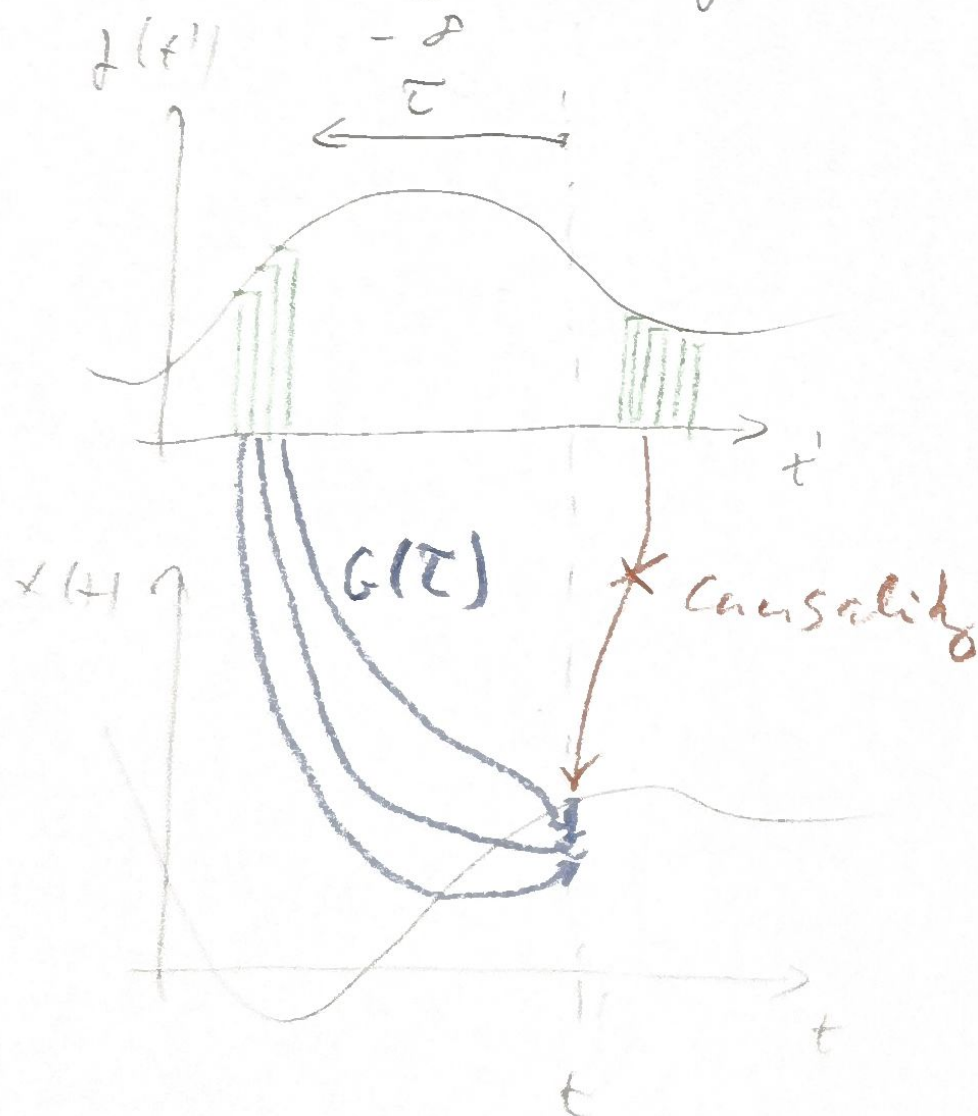
$$x(t) = \int_0^{\infty} G(\tau) f(t-\tau) d\tau$$

Generally we define $G(\tau) := \Theta(\tau) g(\tau)$

to ensure (2) and

$$x(t) = \int_{-\infty}^{\infty} G(\tau) f(t-\tau) d\tau$$

↖ convolution



Calculation of $G(\tau)$

Dirac delta drive: $f(t') := \delta(t')$

$$x(t) = G(t) = \int_{-\infty}^{\infty} G(\tau) \delta(t-\tau) d\tau$$

$$\frac{dG(t)}{dt} + \frac{1}{2} G(t) = \delta(t) \quad G(t) = 0 \text{ for } t < 0$$

$$\int_{-\epsilon}^{\epsilon} \left(\frac{dG(t)}{dt} + \frac{1}{2} G(t) \right) dt = \int_{-\epsilon}^{\epsilon} \delta(t) dt$$

$$G(t=\epsilon) - \underbrace{G(t=-\epsilon)}_{=0} + \underbrace{\frac{1}{2} G(0)\epsilon}_{\rightarrow 0} = 1$$

$$G(t=0) = 1$$

For second order differential eq:

$$G'(t=0) = 1 \quad G''(t=0) = 0$$

Green's function in Fourier space

$$(i) \quad \mathcal{F}[(f * g)(t)] = \mathcal{F}\left[\int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau\right] = \sqrt{2\pi} \hat{f}(\omega) \hat{g}(\omega)$$

convolution

$$\Rightarrow \tilde{x}(\omega) = \sqrt{2\pi} \hat{G}(\omega) \hat{f}(\omega)$$

Differential operator D

$$D(x) = f(t)$$

$$D(G) = f(t)$$

$$D(\tilde{G}) = \frac{1}{\sqrt{2\pi}}$$

Example : damped harmonic oscillator

$$\ddot{x} + 2\alpha \dot{x} + \omega_0^2 x = f(t)$$

$$\mathcal{F}[\dot{x}] = i\omega \tilde{x}$$

$$-\omega^2 \tilde{x} + 2\alpha i\omega \tilde{x} + \omega_0^2 \tilde{x} = \tilde{f}(\omega)$$

$$\tilde{x}(\omega) = \sqrt{2\pi} \tilde{G}(\omega) \tilde{f}(\omega)$$

$$G(\omega) = \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{-\omega^2 + 2\alpha i\omega + \omega_0^2}$$

This can be obtained directly from the diff. eq. using Dirac delta drive

$$-\omega^2 \tilde{G} + 2\alpha i\omega \tilde{G} + \omega_0^2 \tilde{G} = \frac{1}{\sqrt{2\pi}}$$

$$\tilde{G} = \frac{1}{\sqrt{2\pi}} \frac{1}{-\omega^2 + 2\alpha i\omega + \omega_0^2}$$

Euler form

$$\hat{G}(\omega) = |G(\omega)| \cdot e^{-i\phi}$$

$$\sqrt{2\pi} \tilde{G}(\omega) = \frac{(\omega_0^2 - \omega^2) - i2\alpha\omega}{(\omega_0^2 - \omega^2)^2 + 4\alpha^2\omega^2}$$

$$\tan \phi = - \frac{2\alpha\omega}{\omega_0^2 - \omega^2}$$

$$|\tilde{G}(\omega)| = \frac{1}{2\pi \sqrt{(\omega_0^2 - \omega^2)^2 + 4\alpha^2\omega^2}}$$

$$\omega_R = \sqrt{\omega_0^2 - 2\alpha^2}$$

